Physics Department
Fisika Departement

University of Pretoria

Semester Test 1 *Semestertoets 1*

FSK116

15 March 2013 / *15 Maart 2013*

Surname and initials *Van en voorletters*	MEMO	Student number *Studentenommer*	
Signature *Handtekening*			
Time *Tyd*	2½ hours *2½ ure*	Max. Total *Maks. Totaal*	65 Marks Max *65 punte Maks*
Internal Examiner: *Interne Eksaminator*	Dr J M Nel	External Examiner *Eksterne Eksaminator*	Mr R Q Odendaal

• Answer all questions. • Cleary show all calculations. Do not just write down the answer. • Clearly state all assumptions, rules and laws used. • Where applicable draw the free-body diagrams. • Demonstrate clearly that you understand the physics that you use to answer the question. This means that a numerically correct answer will not necessarily be awarded full marks. • Programmable calculators may NOT be used. • No textbooks allowed. • A formula sheet is attached. • All test and examination regulations of the University of Pretoria are applicable.	• *Beantwoord alle vrae.* • *Wys al jou berekeninge. Moenie net 'n antwoord neerskryf nie.* • *Stel duidelik alle aannames, reëls en wette wat u gebruik.* • *Waar van toepassing, teken die vryliggaamsdiagram (of kragte diagramme).* • *Wys dat jy die fisika wat jy gebruik om die vraag te beantwoord verstaan. Dit beteken dat volpunte sal nie noodwendig vir 'n numeries-korrekte antwoord gegee word nie.* • *Programmeerbare sakrekenaars MAG NIE gebruik word nie.* • *Geen handboeke word toegelaat nie.* • *'n Formuleblad is aangeheg.* • *Alle toets- en eksamenregulasies van die Universiteit van Pretoria is van toepassing.*

✂---✂---✂---

Semester Test 1 **FSK 116** **Semestertoets 1**

15 March 2013 / *15 Maart 2013*

Surname and initials *Van en voorletters*		Student number *Studentenommer*	
Signature *Handtekening*		Venue *Lokaal*	

Circle: Circumference $=2\pi r$ = omtrek Area $=\pi r^2$ Sphere: Area $=4\pi r^2$ Volume $=\frac{4}{3}\pi r^3$ Cylinder / Silinder Area $=2(\pi r^2)+2\pi rh$ Volume $=\pi r^2 h$ Trapesium: Area $=\frac{1}{2}(a+b)h$ $\sin^2\theta+\cos^2\theta=1$ $\sin 2\theta = 2\sin\theta\cos\theta$ $\sin(\alpha\pm\beta)=\sin\alpha\cos\beta\pm\cos\alpha\sin\beta$ $\cos(\alpha\pm\beta)=\cos\alpha\cos\beta\mp\sin\alpha\sin\beta$ $\sin\alpha\pm\sin\beta=2\sin\frac{1}{2}(\alpha\pm\beta)\cos\frac{1}{2}(\alpha\mp\beta)$ $\vec{a}\cdot\vec{b}=ab\cos\phi$ $\vec{a}\cdot\vec{b}=a_xb_x+a_yb_y+a_zb_z$	$\vec{a}\times\vec{b}=(a_yb_z-b_ya_z)\hat{i}$ $+(a_zb_x-b_za_x)\hat{j}$ $+(a_xb_y-b_xa_y)\hat{k}$ $x=\dfrac{-b\pm\sqrt{b^2-4ac}}{2a}$ $\Delta L = L\alpha\Delta T$ $\Delta V = V\beta\Delta T$ $Q=C(T_f-T_i)$ $Q=cm(T_f-T_i)$ $Q=Lm$ $W=\int_{V_i}^{V_f} pdV$	$T_c=T_K-273.15°$ $T_F=\frac{9}{5}T_C+32$ $v_{avg}=\frac{1}{2}(v_0+v)$ $v=v_0+at$ $x=x_0+v_0t+\frac{1}{2}at^2$ $x=x_0+v_{avg}t$ $v^2=v_o^2+2a(x-x_0)$ $x-x_0=\frac{1}{2}(v_0+v)t$ $x-x_0=vt-\frac{1}{2}at^2$

Question 1 **6**

Convert the following to SI units:

(a) 1.00 day [3]

1.00 day	$1.00\,\text{day}=\dfrac{24\,\text{hours}}{1\,\text{day}}\times\dfrac{60\,\text{min}}{1\,\text{hour}}\times\dfrac{60\,\text{s}}{1\,\text{min}}$ ✓✓ $=86400\,\text{s}=8.64\times10^4\,\text{s}$ ✓

(b) 80 km/h [3]

80 km/h	$80\,\text{km/h}=\dfrac{80000\,\text{m}}{\text{h}}\times\dfrac{1\,\text{h}}{60\,\text{min}}\times\dfrac{1\,\text{min}}{60\,\text{s}}$ ✓✓ $=22.2\,\text{m/s}$ ✓

IT IS ESSENTIAL THAT YOU INCLUDE YOUR UNITS IN YOUR MULTIPLICATION. YOU CANNOT JUST MULTIPLY NUMBERS.

Question 2 **7**

Vector $\vec{A}$ is directed along the +*x* axis and vector $\vec{B}$ has a magnitude of 6.0 m. The sum of these two vectors is a third vector that is directed along the +*y* axis, with a magnitude 3.0 times that of $\vec{A}$. What is that magnitude of $\vec{A}$?

Vektor $\vec{A}$ is langs die +x-as gerig en vektor $\vec{B}$ het 'n grootte van 6.0 m. Die som van die twee vektore is 'n derde vektor wat langs die +y-as gerig is met 'n grootte van 3.0 keer dié van $\vec{A}$. Wat is die grootte van $\vec{A}$? [7]

Given information: $\vec{A} = a_x\,\hat{\mathrm{i}}$ $\vec{B} = b_x\,\hat{\mathrm{i}} + b_y\,\hat{\mathrm{j}}$ $\lvert\vec{B}\rvert = 6.0\,\mathrm{m} = \sqrt{b_x^2 + b_y^2}$ $\vec{C} = \vec{A} + \vec{B} = (a_x + b_x)\hat{\mathrm{i}} + b_y\,\hat{\mathrm{j}}$ $= b_y\,\hat{\mathrm{j}}$ $\lvert\vec{A} + \vec{B}\rvert = 3.0\lvert\vec{A}\rvert$	
Therefore $a_x = -b_x$ and $b_y = 3a_x$ $\lvert\vec{C}\rvert = \sqrt{b_y^2} = 3.0\left(\sqrt{a_x^2}\right)$ therefore: $\lvert b_y\rvert = 3\lvert a_x\rvert = 3\lvert b_x\rvert$ $\lvert\vec{B}\rvert = 6.0\,\mathrm{m} = \sqrt{b_x^2 + b_y^2}$ $= \sqrt{b_x^2 + (3b_x)^2} = \sqrt{10b_x^2}$ $b_x^2 = \frac{36}{10} = 3.6$ $\lvert b_x\rvert = 1.90\ \mathrm{m} = \lvert a_x\rvert$ $\lvert\vec{A}\rvert = 1.9\ \mathrm{m}$	✓ ✓ ✓ ✓ ✓ ✓ ✓

Question 3 **7**

In the product $\vec{F} = q\vec{v}\times\vec{B}$, take q = 3.0 C, $\vec{v} = 2.0\,\hat{\imath} + 4.0\,\hat{\jmath} + 6.0\,\hat{k}$ m.s^{-1} and $\vec{F} = 4.0\,\hat{\imath} + -2.0\,\hat{\jmath} + 12\,\hat{k}$ N. Determine $\vec{B}$ (in tesla (T)) in unit vector notation if $B_x = B_y$.

In die produk $\vec{F} = q\vec{v}\times\vec{B}$, *aanvaar q = 3.0 C,* $\vec{v} = 2.0\,\hat{\imath} + 4.0\,\hat{\jmath} + 6.0\,\hat{k}$ *m.s*$^{-1}$ *en* $\vec{F} = 4.0\,\hat{\imath} + -2.0\,\hat{\jmath} + 12\,\hat{k}$ *N. Bepaal* $\vec{B}$ *(in tesla (T)) in eenheid-vektor notasie as* $B_x = B_y$. [7]

$\vec{F} = q\vec{v}\times\vec{B}q\left(v_x\,\hat{\imath} + v_y\,\hat{\jmath} + v_z\,\hat{k}\right)\times\left(B_x\,\hat{\imath} + B_y\,\hat{\jmath} + B_z\,\hat{k}\right)$	
$= q\left(\left(v_y B_z - v_z B_y\right)\hat{\imath} + \left(v_z B_x - v_x B_z\right)\hat{\jmath} + \left(v_x B_y - v_y B_x\right)\hat{k}\right)$	✓✓✓
$= 4.0\,\hat{\imath} + -2.0\,\hat{\jmath} + 12\,\hat{k}$	
Therefore:	
$F_x = q\left(v_y B_z - v_z B_y\right) = 4.0$	
$F_y = q\left(v_z B_x - v_x B_z\right) = -2.0$	
$F_z = q\left(v_x B_y - v_y B_x\right) = 12$	
If $B_x = B_y$, then	
$F_z = 3.0\left(2.0B_x - 4.0B_x\right) = 12N$	✓
$-6.0B_x = 12$	
$B_x = -2.0\,\text{T} = B_y$	✓
And due to an error in the question, the answer was marked depending of whether you used Fy or Fx to solve for Bz.	
Final answer had to be in unit vector notation:	✓
$\vec{B} = -2.0\,\hat{\imath} - 2.0\,\hat{\jmath} - (B_z value)\,\hat{k}$	✓

You need to multiply the cross product out so that you have the three components of the force vector in terms of the magnetic field components.

Question 4 **6**

An aluminium-alloy rod has a length of 10.000 cm at 20.000 °C and a length of 10.020 cm at the boiling point of water.

'n Aluminium-allooi staaf het 'n lengte van 10.000 cm by 20.000 ° C en 'n lengte van 10.020 cm by die kookpunt van water.

(a) What is the length of the rod at 50.000 °C?

Wat is die lengte van die staaf by 50.000 °C? [3]

Given: $\Delta T = 80.000$ °C $\quad L_o = 10.000$ cm $\Delta L = 0.020$ cm From the definition of expansion: $\Delta L = 0.020\,\text{cm}$ $= L\alpha\Delta T$ $\alpha = \frac{\Delta L}{L\Delta T} = \frac{0.020\,\text{cm}}{10.000\,\text{cm}(80\,\text{K})}$ $= 25.000\times10^{-6}\,/\,\text{C}°$	You need to first determine α ✓
$\Delta L_{50} = L\alpha\Delta T$ $= 10.000\,\text{cm}\left(25.000\times10^{-6}\,/\,C°\right)(30\,\text{K})$ $= 75.000\times10^{-4}$ cm $L = 10.0075$ cm at 50°C	✓ ✓

(b) What is the temperature of the rod if its length is 10.003 cm?

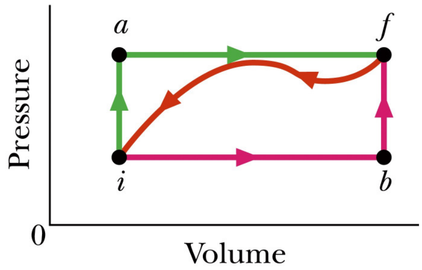
Wat is die temperatuur van die staaf indien die lengte 10.003 cm is? [3]

Given: $\Delta T = 80.000$ °C $\quad L_o = 10.000$ cm $\Delta L = 0.003$ cm $\Delta L = 0.003\,\text{cm}$ $= L\alpha\Delta T$ $\Delta T = \frac{\Delta L}{L\alpha} = \frac{0.003\,\text{cm}}{10.000\,\text{cm}\left(25.000\times10^{-6}\,/\,\text{C}°\right)}$ $= 12\,°\text{C}$ $T = 32$°C	✓✓please use α from part (a) ✓ ✓

Question 5 **9**

When a system is taken from state i to state f along path *iaf* in the figure shown, Q = 210 J and W = 84 J. Along path *ibf* Q = 150 J.

Wanneer 'n stelsel van toestand i na toestand f geneem word langs die pad iaf soos aangedui in die figuur is Q = 210 J en W = 84 J. Langs pad ibf is Q = 150 J.



(a) Determine W along path *ibf*.
Bepaal W langs pad ibf. [3]

For path *iaf*: $\Delta E_{\text{int}} = Q - W = 210\,\text{J} - 84\,\text{J} = 126\,\text{J}$ This is the same for all paths *i-f*. For path *ibf*: $\Delta E_{\text{int}} = Q - W$ $W = Q - \Delta E_{\text{int}} = 150\,\text{J} - 126\,\text{J}$ $= 24\,\text{J}$	✓ ✓ ✓

(b) If $W = -54$ J for the return path *fi*, determine Q for this path.
Indien $W = -54$ J *vir die pad fi, bepaal Q vir hierdie pad.* [3]

For path *fi*: $\Delta E_{\text{int}} = Q - W$ $Q = \Delta E_{\text{int}} + W = -126\,\text{J} + (-54\,\text{J})$ $= -180\,\text{J}$	✓✓negative ΔE !and W ✓

(c) If $E_{int,i}$ = 41 J, what is $E_{int,i}$?
Indien $E_{int,i}$ *= 41 J, wat is* $E_{int,i}$*?* [3]

From (a) and path *iaf:* $\Delta E_{\text{int}} = Q - W = 210\,\text{J} - 84\,\text{J} = 126\,\text{J}$ Thus $\Delta E_{\text{int}} = E_{\text{int},f} - E_{\text{int},i}$ $E_{\text{int},f} = \Delta E_{\text{int}} + E_{\text{int},i}$ $= 126\,\text{J} + 41\,\text{J}$ $= 167\,\text{J}$	✓ ✓ ✓

Question 6 **7**

When the acceleration is constant, the average and instantaneous acceleration are equal:

$a = a_{avg} = \frac{v - v_0}{t - 0}$, where a is the acceleration, v is the velocity at time t and v_o is the initial velocity at $t = 0$ s. We can also write the average velocity as $v_{avg} = \frac{x - x_0}{t - 0}$, where x is the position at time t and x_o is the initial position at $t = 0$ s. Using these two equations show that when a particle is accelerating in the x direction, its velocity is given by $v^2 = v_0^2 + 2a(x - x_o)$. Show all steps and state all reasoning clearly.

Wanneer die versnelling konstant is, is die gemiddelde en oombliklike versnelling dieselfde:

$a = a_{gem} = \frac{v - v_0}{t - 0}$ *waar a die versnelling is, v die snelheid by tyd t en* v_o *die aanvanklike snelheid by t = 0 s is. Ons kan ook die gemiddelde snelheid as* $v_{gem} = \frac{x - x_0}{t - 0}$ *skryf, waar x die posisie by tyd t en* x_o *die aanvanklike posisie by t = 0 s is. Deur gebruik te maak van hierdie twee vergelykings, bewys dat* $v^2 = v_0^2 + 2a(x - x_o)$. *Wys al die stappe en verduidelik jou aannames.*

[7]

$a = a_{avg} = \frac{v - v_0}{t - 0}$ ① $v = v_0 + at$	✓
$v_{avg} = \frac{x - x_0}{t - 0}$ ② $x = x_0 + v_{avg} t$	✓
But $v_{avg} = ½(v + v_0)$.	✓
Substituting this for v_{avg} in ② : $x = x_0 + \left[½(v + v_0)\right]t$	✓
From the first equation, we substitute for v and rearrange the terms: $x = x_0 + \left[½(v + v_0)\right]t$ $x - x_0 = ½vt + ½v_0 t$	✓
$= ½(v_0 + at)t + ½v_0 t$ $= v_0 t + ½at^2$	
From ①: $t = \frac{v - v_0}{a}$	✓
Substituting into above:	

$x - x_0 = v_0 t + ½at^2$ $= v_0\left(\frac{v - v_0}{a}\right) + ½a\left(\frac{v - v_0}{a}\right)^2$ $2(x - x_0)a = 2v_0(v - v_0) + (v - v_0)^2$ $2a\Delta x = 2v_0 v - 2v_0^{\ 2} + v^2 + v_0^{\ 2} - 2v_0 v$ $v^2 = v_0^{\ 2} + 2a(x - x_0)$	✓
$a = a_{avg} = \frac{v - v_0}{t - 0}$ ① $v_{avg} = \frac{x - x_0}{t - 0}$ ② $x = x_0 + v_{avg} t$ From ①: $t = \frac{v - v_0}{a}$ ③ Substituting ③ in ②: $\frac{(x - x_0)a}{v - v_0} = ½(v + v_0)$ $2(x - x_0)a = (v + v_0)(v - v_0)$ $= v^2 - v_0^2$ $v^2 = v_0^2 + 2a(x - x_0)$	**Shorter method**

[3]

Question 7 **9**

A motorcyclist who is moving along an x axis directed toward the east has an acceleration given by $a = (6.1 - 1.2t)$ m.s^{-2} for $0 \le t \le 6.0$ s. At t = 0 s, the velocity and position of the cyclist are 2.47 m.s^{-1} and 7.3 m respectively.

'n Motorfietsryer beweeg oos langs 'n x-as gerig en het 'n versnelling gegee deur $a = (6.1 - 1.2t)\,ms^{-2}$ vir $0 \le t \le 6.0$ s. By t = 0 s, is die die snelheid en posisie van die fietsryer 2,47 ms^{-1} en 7,3 m onderskeidelik.

(b) What is the maximum speed achieved by the cyclist?
Wat is die maksimum spoed wat die motorfietsryer bereik? [5]

$a = (6.1 - 1.2t)$ The velocity will be maximum when the first derivative of v ie. acceleration = 0 m/s^2.	
$a = (6.1 - 1.2t) = 0$ $t = \frac{6.1}{1.2} = 5.08 \approx 5.1\,\text{s}$ The velocity is obtained by the integrating the acceleration expression	✓ ✓

$a=(6.1-1.2t)$ $v=\int a dt=\int(6.1-1.2t)dt$ $=6.1t-0.6t^2+c$ Since v = 2.47 m/s at t = 0 s, then c = 2.47 m/s Need to now determine the maximum speed at 5.1 s $v=\left[6.1(5.1)-0.6(5.1)^2+2.47\right]$ $=17.97\,\mathrm{m.s^{-1}}$	✓ ✓ ✓

(b) What total distance does the cyclist travel between t = 0 s and 6.0 s.
Wat is die totale afstand afgelê deur die ryer tussen t = 0 s en 6.0 s. [4]

Total distance will be the area under the v-t curve (integral of the $v(t)$ function between 0 and 6 s. $v=6.1t-0.6t^2+2.47$ $x=\int v dt=\int_0^6(6.1t-0.6t^2+2.47)dt$ $=\left[3.05t^2-0.2t^3+2.47t\right]_0^6$ $=81.4$ m You need to determine the x value (total distance) between the limits of t = 0s and t = 6s, since the starting point at t = 0s is at x = 7.3 m.	✓ ✓✓ ✓
The following is not necessary, but could have been used to determine distance travelled. $x=\int v dt=\int_0^6(6.1t-0.6t^2+2.47)dt$ $=\left[3.05t^2-0.2t^3+2.47t+c\right]$ Since x = 7.3 m/s at t = 0 s, then c = 7.3 m $x_f=\left[3.05(6)^2-0.2(6)^3+2.47(6)+7.3\right]$ $\quad x_i=7.3$ m $=109.8-43.2+14.8+7.3$ $=88.7\,\mathrm{m}$ $\Delta x=x_f-x_i$ $=88.7-7.3=81.4\,\mathrm{m}$	

Question 8 **8**

A small rocket, such as those used for meteorological measurements of the atmosphere, is launched vertically with an acceleration of 30 m/s^2. It runs out of fuel after 30 s. What is the maximum altitude of the rocket?

'n Klein vuurpyl, soos dié wat gebruik word vir weerkundige metings van die atmosfeer, word vertikaal gelanseer met 'n versnelling van 30 m/s^2. Die brandstof raak op na 30 s. Wat is die maksimum hoogte van die vuurpyl? [8]

There are two stages to this problem. Firstly, when the rocket is firing and using fuel, secondly when the fuel is finished, the rocket is then in free-fall.

Phase 1 (before fuel runs out):

v_o = 0 m/s a_1 = +30m/s^2 t = 30 s y_1 = ?

v_1 = ?

$$v_1 = v_0 + a_1 t$$
$$= (30\,\mathrm{m.s^{-2}})(30\,\mathrm{s}) \quad ✓✓$$
$$= 900\,\mathrm{m.s^{-1}}$$

$$y_1 = y_0 + v_0 t + \tfrac{1}{2} a_1 t^2$$
$$= 0\,\mathrm{m} + (0\,\mathrm{m.s^{-1}})(30\,\mathrm{s}) + \tfrac{1}{2}(30\,\mathrm{m.s^{-2}})(30\,\mathrm{s})^2 \quad ✓✓$$
$$\Delta y_1 = 13500\,\mathrm{m}$$

The rocket is moving upwards at a velocity of 900 m/s. It has not yet reached its maximum height. Phase 2 takes care of the distance travelled till vy = 0 m/s.

Phase 2 (after fuel runs out):

v_1 = 900 m/s a_1 = - 9.8m/s^2 y_2 = ?

v_2 = 0 m/s✓

$$v_2^2 = v_1^2 + 2a_2\Delta y$$
$$\Delta y = \frac{v_2^2 - v_1^2}{2a_2}$$
$$= \frac{(0\,\mathrm{m.s^{-1}})^2 - (900\,\mathrm{m.s^{-1}})^2}{2(-9.8\,\mathrm{m.s^{-2}})} \quad ✓$$
$$= 41326\,\mathrm{m}$$

$$\Delta y = \Delta y_1 + \Delta y_2 = 13.5\times10^3\,\mathrm{m} + 41.3\times10^3\,\mathrm{m} \quad ✓✓$$
$$= 54.8\times10^3\,\mathrm{m}$$

ENJOY THE WEEKEND / GENIET DIE NAWEEK